



Effect of magnetic field on diffusion of ethylammonium nitrate – water mixtures confined between polar glass plates

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ARTICLE INFO

Article history:

Received 20 August 2018

Received in revised form 11 October 2018

Accepted 16 October 2018

Available online 18 October 2018

Keywords:

Nuclear magnetic resonance

Diffusivity

Ion dynamics

Water diffusion

ABSTRACT

We used ¹H NMR diffusometry to study mixtures of ethylammonium nitrate (EAN) with water (3.1–12.4 mol% of added H₂O) confined between polar glass plates and exposed to a static magnetic field of 9.4 T. The presence of such restrictions reverses the concentration dependence of the diffusivities of the EA (ethylammonium) cation and water typical for the bulk system. The presence of water weakens the effects of a static magnetic field on diffusion of the EA cation as well as on proton exchange of –NH₃ groups. Surprisingly, the amplitude of the echo signal of water protons decreases during exposure to the magnetic field and finally disappears, a phenomenon that depends on the concentration of water in the system. Based on experimental data, we suggest that water in the system is present in two states with different dynamic properties. One type of water formed in confinement possesses NMR relaxation time typical for liquids; its diffusivity can be measured by ¹H NMR. The second type of water is formed upon exposure of the sample of the first type to the magnetic field and eventually includes all the water in the system. This type of water possesses “solid-like” NMR relaxation features that makes it “invisible” to the NMR diffusometry technique. We suggest that this second type of water is adsorbed onto the glass plates. Correspondingly, EAN exists in two liquid phases: the first one contains an EAN–water mixture, while the second one contains neat EAN, and forms on the microscopic scale range under the influence of a static magnetic field.

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1. Introduction

Ionic liquids (ILs) are a new class of designed electrolytes, solvents, catalysts and CO₂ absorbents [1–4]. They possess properties of interest for industrial applications, such as a broad range of temperatures for the liquid state, high thermal stability, low flammability and good miscibility with water and organic liquids [5–9]. ILs that are synthesized by the neutralization of the reaction of a Brønsted acid and a Brønsted base have exchangeable protons and are called protic ILs (PILs) [1]. Ethylammonium nitrate (EAN), the most frequently reported PIL [1], is used as a medium for chemical reactions, as a precipitating agent for protein separation [10], and as an electrically conductive solvent in electrochemistry [11]. Mixing ILs with solvents modifies their transport and thermodynamic properties [5,12]. Particularly, EAN exhibits important hygroscopic and hydrophilic characteristics, which allow it to mix with water and alcohols at any proportion [5]. It has been shown that solvation of the EAN ions by water molecules weakens the strong

Coulombic and hydrogen bond interactions between ions of opposite charge, leading to a decrease in viscosity [5]. X-ray spectra of neat EAN were consistent with its nanoscale structure and essentially independent of increased water content [13]. Nano-heterogeneity in the form of a bi-continuous nanostructure has been observed in EAN and EAN–D₂O systems [14]. Hayes et al. [7] proposed a micelle-like model for the solvation structure of EAN–water, with water molecules located in the bulk polar domains where they are associated with the charge groups of the ions. The local structure of the polar domains closely resembles pure water and pure EAN, confirming that these interactions of water with ions are unaffected by mixing. It has been suggested that water increases the effective head group size of the cation, which raises the interfacial curvature around the non-polar regions [7]. This changes the EAN nanostructure from a near-zero mean curvature, or locally flat L₃-sponge-like in pure EAN, to a branched (locally cylindrical) network or mesh in EAN/water mixtures [7].

In various technical applications, ILs are used in nano-scale and micro-meter scale confinements [15–17], which also can change the transport properties of the ILs [18–21]. Confined ILs [17,20–23], particularly EAN [21–24], have attracted special interest during the last few years. Enhanced diffusion of ethylammonium (EA) cations has been

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observed for EAN confined between polar glass plates [21]. It has been experimentally established that self-diffusion and NMR relaxation of EAN confined between polar glass plates reversibly alters after placement in a strong static magnetic field [23]. This effect can provide a new tool to manipulate and fine-tune the properties of ionic liquids for various technological applications (lubrication, electrochemistry, supported catalysis, peptide separation) and also expand their applications into new fields. In addition, this effect can alter the properties of ILs in thin layers, which is one of the possible new applications in portable electronic devices.

The main factors responsible for this effect are the availability of protons in the protic EAN and the polarity of the surface; the exchange rate of mobile protons between $(-\text{NH}_3)^+$ and $(\text{NO}_3)^-$ chemical groups of EAN plays a crucial role in the observed processes. It has been argued that the processes influencing the dynamics of EAN in such confinement are the phase transformations of EAN [23,24]. The process can be described well by the Avrami equation that is typical for autocatalytic processes [24]. Exposure to the magnetic field also leads to a decrease in the ^1H NMR signal of residual water, which is most probably related to the adsorption of water on the polar surfaces of the glass plates [24]. The role of water or any other solvent in the dynamics of confined ILs exposed to a magnetic field has not yet been studied. NMR diffusometry is a very convenient method to study the translational dynamics of ions and additions in ionic liquids [3,4,12,18,20,21,23–27].

In this work, the diffusion of EAN ionic liquid in systems confined between polar glass plates exposed to a static magnetic field is further investigated by ^1H NMR diffusometry. Mixtures of EAN-water with low concentrations of water were the subject of this study.

2. Materials and methods

2.1. Sample preparation

Structures of ions of ethylammonium nitrate are shown in Fig. 1. EAN was produced and characterized at the Chemistry of Interfaces of Luleå University of Technology as described previously [21]. Despite the common use of a vacuum step to dry the ILs, some traces of water can still remain [28]. The remaining water becomes strongly hydrogen bonded or ionized in the ILs and, thus, is less likely to be removed compared to some other species. The quantity of water in the synthesized EAN was less than 0.055 wt% as determined by Karl-Fisher titration (Metrohm 917 Karl Fisher Coulometer with HYDRANAL reagent).

Double distilled water (H_2O) was used to prepare a mixture of EAN- H_2O with added weight concentrations of water at 0.5, 1.0 and 2.0%. These concentrations correspond to 3.1, 6.2 and 12.4 mol%, respectively. EAN and water were mixed in a plastic 1.5-ml Eppendorf tube and left for a week for equilibration.

NMR measurements for the bulk EAN and its mixtures with water were performed by placing 300 μL of the IL in a standard 5-mm NMR glass tube. Confined EAN mixtures were prepared with glass plates arranged in a stack (see Fig. S1 in the Electronic Supplementary information [ESI]). The plates ($14 \times 2.5 \times 0.1$ mm, Thermo Scientific Menzel-Gläser, Menzel GmbH, Germany) were carefully cleaned before sample preparation. Contact angle measurements with Milli-Q water gave a

contact angle near 0° . A stack of glass plates filled with IL was prepared in a glove box in a dry N_2 atmosphere. Samples were made by adding 2 μL of EAN mixture to the first glass plate, placing a new glass plate on top, adding 2 μL on top of this glass plate, etc. until the thickness of the stack reached 2.5 mm. IL that overflowed at the edges of the stack was removed by sponging. The sample consisting of a stack of ca 37 glass plates was placed in a rectangular glass tube with further sealing. The mean spacing between the glass plates was assessed as reported earlier [21], which was determined to be $d \sim 3.8\text{--}4.5$ μm . A detailed report of the sample preparation and characterization has been described in our previous papers [21,23]. To allow equilibration of IL inside the sample, experiments were started a week after sample preparation.

2.2. Pulsed-field gradient diffusometry

^1H NMR self-diffusion measurements were performed using a Bruker Ascend/Aeon WB 400 (Bruker BioSpin AG, Fällanden, Switzerland) NMR spectrometer with a working frequency of 400.27 MHz for ^1H , magnetic field strength of 9.4 T with magnetic field homogeneity better than $4.9 \cdot 10^{-7}$ T, using a Diff50 Pulsed-Field-Gradient (PFG) probe.

The diffusional decays (DD) were recorded using stimulated echo pulse sequences. For single-component diffusion, the form of DD can be described as [27]:

$$A(\tau, \tau_1, g, \delta) \propto \exp\left(-\frac{2\tau}{T_2} - \frac{\tau_1}{T_1}\right) \exp\left(-\gamma^2 \delta^2 g^2 D t_d\right). \quad (1)$$

Here, A is the signal intensity, τ and τ_1 are the time intervals in the pulse sequence; γ is the gyromagnetic ratio for protons; g and δ are the amplitude and the duration of the gradient pulse; $t_d = (\Delta - \delta/3)$ is the diffusion time; $\Delta = (\tau + \tau_1)$; D is the diffusion coefficient. In the measurements, the duration of the 90° pulse was 7 μs , δ was in the range of 0.5–2 ms, τ was in the range of 3–5 ms, and the amplitude of g was varied from 0.06 up to 29.73 $\text{T} \cdot \text{m}^{-1}$. The recycle delay was 3.5 s.

As has been shown [21], physical restrictions (glass plates) have no effect on the D of the IL in the direction along the plates, while for diffusion normal to the plates, the effect of the plates on D was perceptible at t_d longer than 3 ms. All measurements were performed at 293 K (room temperature), therefore we did not wait until thermal equilibration of the sample in the NMR probe was reached. Measurements were started directly (in 1–2 min) after placing the sample in the NMR probe, tuning and matching the probe. The time required for a single NMR diffusion measurement was around 40 s. Data were processed using Bruker Topspin 3.5 software.

^1H T_1 and T_2 NMR relaxation time measurements were performed with inversion-recovery ($180^\circ\text{--}\tau\text{--}90^\circ\text{--fid}$) and CPMG ($90^\circ\text{--}\tau\text{--}180^\circ\text{--}\tau\text{--echo}$) pulse sequences, respectively [29]. The duration of 90° pulse was 7 μs . Phases of 90° and 180° pulses as well as the phase of the receiver in the CPMG pulse sequence were varied with a specific cycle, and the echo time was fixed to allow elimination of diffusion and J -coupling modulation effects. All measured NMR relaxation times in liquid phase were longer than 20 ms. Therefore, all protons present in the liquid phase of the EAN-water mixture contribute to the stimulated echo of the diffusion NMR experiment.

3. Results and discussion

3.1. Diffusivity

3.1.1. Diffusivity in bulk EAN- H_2O mixtures

The ^1H NMR spectrum of EAN-water mixture demonstrates resonance lines corresponding to the $-\text{NH}_3$, $-\text{CH}_2-$ and $-\text{CH}_3$ groups of EAN, and water protons (Fig. 2) in accordance with our earlier finding (see Electronic Supplementary Information to the reference [21]). The

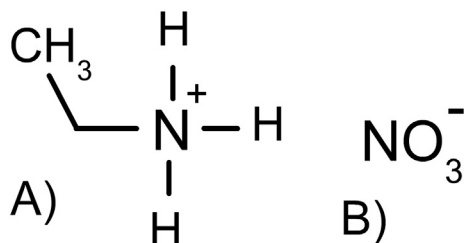


Fig. 1. Chemical structures of ions: A) ethylammonium (EA) cation and B) nitrate anion.

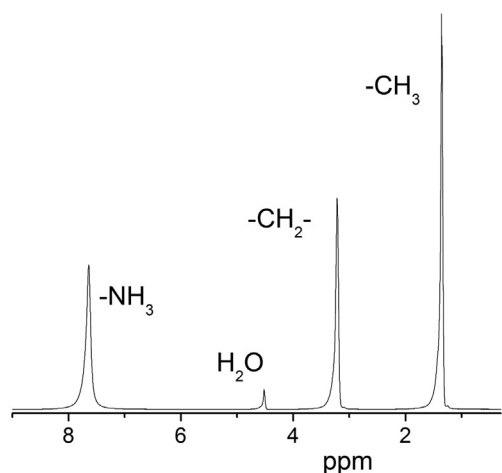


Fig. 2. ^1H NMR spectrum of bulk EAN- H_2O mixture (6.2 mol% H_2O).

intensity of the resonance line corresponding to water increases with increasing water content in the mixture.

Diffusion decays for both, EAN, and water (Fig. 3) are close to single-exponential form (Eq. (1)), the same as for neat bulk EAN [21]. No diffusion time dependence of the diffusion coefficients was found in the range of the diffusion time used in the experiments, 3–300 ms.

Values of D_s for EAN and water in bulk mixtures are shown in Fig. 4 as dotted lines. From this figure, one can see that the diffusion coefficient of water is *ca.* one order of magnitude larger than D of EAN. Bulk diffusion behavior in EAN-water mixtures is typical for solutions. Indeed, the diffusion coefficient of water is larger than D of EAN in bulk and in confinement; the most obvious reason for this is the larger size of an ethylammonium cation compared to a water molecule. This is typical for ILs as well as for solutions of other substances, such as polymers, with molecular weights larger than the molecular weight of the solvent. In bulk samples increasing the water concentration to 12.4 mol% leads to an increase of the D_s for both, EAN, and water, in agreement with the previous finding of Zarrougui et al. concerning decreased viscosities in these systems [5]. In bulk samples, D_s of both, EAN, and water do not change during exposure to the magnetic field.

3.1.2. Diffusivity of EAN and water in EAN- H_2O mixtures confined between polar glass plates

For EAN- H_2O mixtures confined between polar glass plates, diffusion decays for both, EAN, and water (Fig. S2 in the ESI) were close to the single-exponential form (Eq. (1)), the same as for neat EAN and bulk EAN- H_2O mixtures (Fig. 3). No diffusion time dependence of the

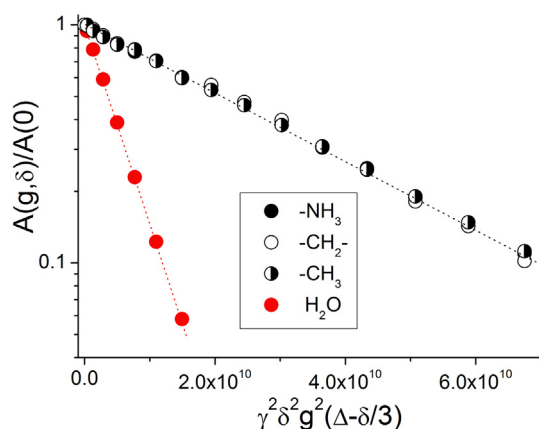


Fig. 3. Diffusion decays for integrals of ^1H NMR resonance lines for bulk EAN- H_2O mixture (6.2 mol% H_2O). $\delta = 1$ ms, $\Delta = 10$ ms.

diffusion coefficients was found at measurements along the glass plates, the same as for neat EAN [21,23]. Typically for liquid mixtures and solutions, diffusion coefficients of constituents increase when the concentration of solvent increases. This is usually explained by the decrease in viscosity of the system [5]. In the bulk EAN-water mixture, D_s of both EAN and water follow the same trend; diffusivities increase when the concentration of solvent (water) in the mixture increases (Fig. 4, dotted lines). On the other hand, diffusional behavior of both, EAN, and water in confinement appears quite unusual. Right after placing the sample in the static magnetic field of $9.4 \text{ T} \cdot \text{m}^{-1}$, values of the diffusion coefficients (D^* s) for EAN for 0, 3.1 and 6.2 mol% of added water in the mixture are larger than the corresponding bulk values D_0 (Fig. 4, solid circles), but the ratio D^*/D_0 decreases from 3.8 (for neat EAN), 2.4 to 1.4, respectively, when more and more water was added in the mixture. For the mixture with 12.4 mol% of water, D^* is smaller D_0 . Values of D^* for water in the confined systems are either larger (for a sample with 3.1 mol% of H_2O) and smaller (for samples with 6.2 and 12.4 mol% of H_2O) than the corresponding bulk values (Fig. 4, open circles). Generally, an increase in the concentration of water in the confined EAN-water system leads to a decrease of D^* s of both EAN and water, a trend that is opposite that observed in bulk EAN-water systems.

The diffusivities of liquids in various confinements have been studied earlier for different liquids introduced in porous and dispersed solids with different geometries and sizes of pores and voids [17,27,30]. In the case of nano-pores, the local diffusion coefficient usually decreases due to the interaction of the liquid molecules with the pore walls [17,30], but in some particular cases, the local D may increase due to decreased packing of the liquid molecules in the restricted volume [18,19]. In the case of macro-pores, the local D usually does not change, because the contribution of molecules interacting with the macro-pore walls is infinitesimal. However, the apparent diffusion coefficient, which is measured at displacements of molecules that much exceed the sizes of the pores, typically decreases as a result of the tortuosity of the porous space [31]. Surprisingly, the diffusion coefficient of EAN confined between polar glass plates just after placing a sample in the NMR spectrometer (D^*) is larger than the corresponding values of D in bulk mixtures in the whole range of water concentrations used in this study. The increase in the diffusion coefficient of EAN (for samples with 0, 3.1 and 6.2 mol% H_2O) and water (for 3.1 mol% H_2O) relative to their values in bulk, observed in our experiment with the distance between glass plates *ca.* $4 \mu\text{m}$, which is in the range of macro-pores (Fig. 4), was not observed earlier for molecular and ionic liquids in any type of confinement. To explain such an unusual effect of EAN D_s , we used different approaches. First, an increase of D of solute and solvent may happen if the mixture decomposed into two liquid phases with larger and smaller concentrations of the solute (particularly, neat solute and solvent phases). This is not expected for the EAN-water system, because EAN is hydrophilic, and it mixes with water at any proportion [5]. Another suggestion is that the structure typical for EAN-water in bulk, which is a sponge-like structure [7], changes in confinement or due to interaction with the polar glass surface. The possibility of such a structural change agrees with the structural polymorphism of the EAN-water mixture [7] and also with our previous studies of the diffusion of EAN confined between glass plates [21,23,24]. The decrease in the diffusivity of water at increasing water concentrations, obviously, is also related to the phase transformation in the EAN-water system occurring in the presence of polar restrictions and at increased water content.

During exposure to the external magnetic field, the diffusivity of EAN in the confined EAN-water system decreased and reached stationary values, D_∞ , which are close to the D_∞ of confined neat EAN and to the bulk value of the diffusion coefficients of neat EAN. This effect has been observed before for confined neat EAN (without added water) [23,24]. Thus, there is evidence that the state, to which the system changes during exposure in the static magnetic field resembles the bulk EAN.

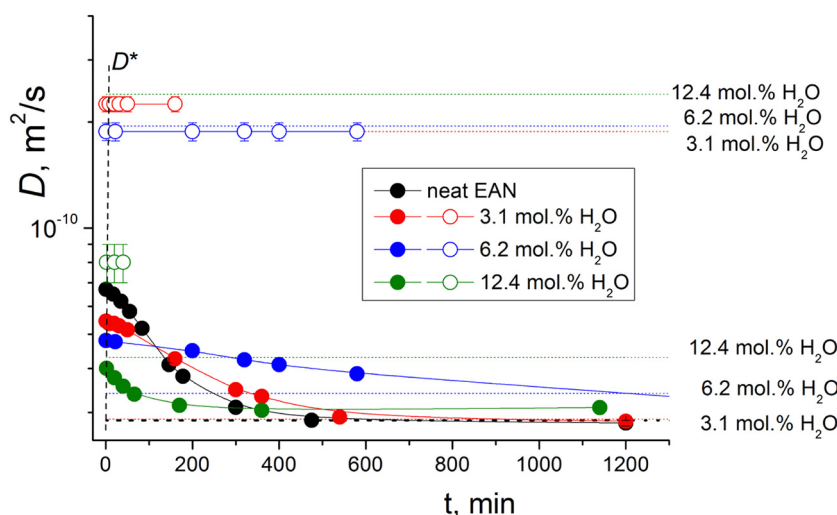


Fig. 4. Dependence of diffusion coefficients of EAN-H₂O mixtures at different times of exposure of the sample to the static magnetic field of 9.4 T. Black dotted line corresponds to EAN, while colored dotted lines with captions on the right correspond to bulk mixtures, while symbols and solid lines correspond to confined systems: EAN (solid circles) and water (open circles). D^* s are diffusion coefficients of EAN and water just after placement in the magnetic field. (Diffusion coefficients for EA⁺ cations in bulk samples of neat EAN and EAN-3.1 mol% H₂O samples are almost identical (see overlapping dotted red and black lines on the bottom of the graph).)

Therefore, a new EAN-water system state is achieved, which is different from both, bulk, and confined measured in the beginning of the experiment. This change occurs with different rates for samples with different water contents (Fig. 4, solid circles). The rate depends non-monotonously on the water concentration: quite slower change occurs for the sample containing 6.2 wt% in comparison with 3.1 wt% and 12.4 wt% of H₂O. Measurements were repeated few times with different samples, and the effect was reproduced.

However, the diffusion coefficients of water in the same process do not change (Fig. 4, open circles). The ¹H NMR echo signal of water protons decreases with exposure of the system to a static magnetic field and finally disappears after ~200 min (3.1 mol% of water), ~600 min (6.2 mol% of water) and ~100 min (12.4 mol% of water).

Just after placement of a sample containing a mixture of EAN-water, the diffusion coefficient of the EA⁺ cation start to change from D^* to D_∞ (Fig. 4). The rate of these changes depends on the water concentration. It was shown earlier that the kinetics of the change for the sample with confined EAN without added water can be described by the Avrami equation [24]. We analyzed the dependence of D on the time of exposure to the static magnetic field from Fig. 4 using the same method as described in the reference [24]. Obtained Avrami plots are shown in the Supplementary material section (Fig. S3). Generally, the kinetics of transformation fit well with the Avrami equation for the EAN-water system. The Avrami exponent n remains close to 1, which suggests that there is no change in the geometry of the growing phase, while the rate parameter K decreases with increased water up to 6.2 mol% and increases again for a sample containing 12.4 mol% water.

After the confined EAN-water samples were removed from the static magnetic field, they recover their diffusivities (both, EAN, and water) as was observed earlier in the case of neat EAN samples [24].

3.2. ¹H NMR spectra of EAN-H₂O mixtures confined between polar glass plates

Exposure to a magnetic field of confined EAN without added water leads not only to a change in the diffusivity of the EA⁺ cation, but to a change in the intensity of the resonance lines in the ¹H NMR spectrum [23]. An example of a typical change in the ¹H NMR resonance lines during exposure of a confined EAN-H₂O sample to a magnetic field is shown in Fig. 5. These spectra correspond to the first point of diffusion decays used to measure diffusion coefficients. The main features of the change

are: (i) lack of change in amplitudes of the -CH₂- and -CH₃ lines, (ii) increase in amplitude of the -NH₃ line and (iii) decrease in amplitude of the H₂O line, which eventually reaches a value of zero. A detailed presentation for the change in the signals for -NH₃ and H₂O in confined EAN-H₂O samples with water concentrations up to 12.4 mol% is shown in Fig. 6. For the sample without added water, the line corresponding to the -NH₃ group is not observed in the beginning of the experiment, in agreement with a previous study [21], while for samples with added water, the resonance line for protons in the -NH₃ group is actually observed and its integral intensity is larger for samples, in which the water content is larger. During exposure to the magnetic field, the amplitude of this resonance line changes more strongly for samples without added water and with low water concentrations (0 and 3.1 mol% of H₂O), see Fig. 6A and B. The NMR resonance line, corresponding to water protons, decreases in intensity during exposure to the magnetic field and finally disappears at sufficiently long exposure.

Because the compositions of each sample do not change during exposure to the magnetic field, it is evident that the alteration in the

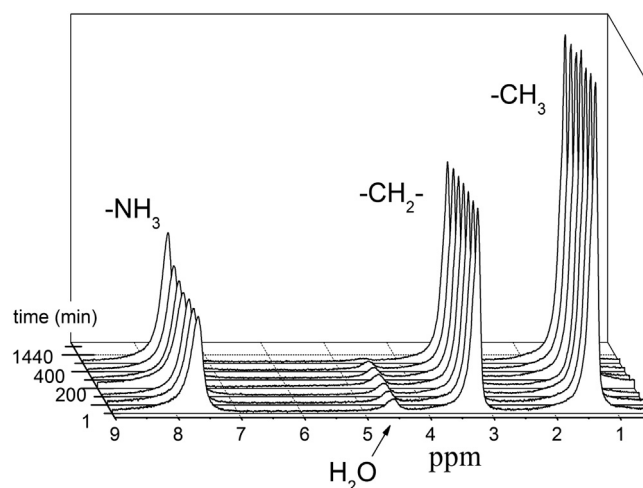


Fig. 5. Typical set of spectra for EAN-H₂O (6.2 mol% H₂O) mixture confined between polar glass plates obtained at different times of exposure of the sample to a magnetic field of 9.4 T. Spectra were acquired by Fourier transformation of spin echo (SE) at $\tau = 5.634$ ms, $g = 0.006$ T/m, $\delta = 1$ ms.

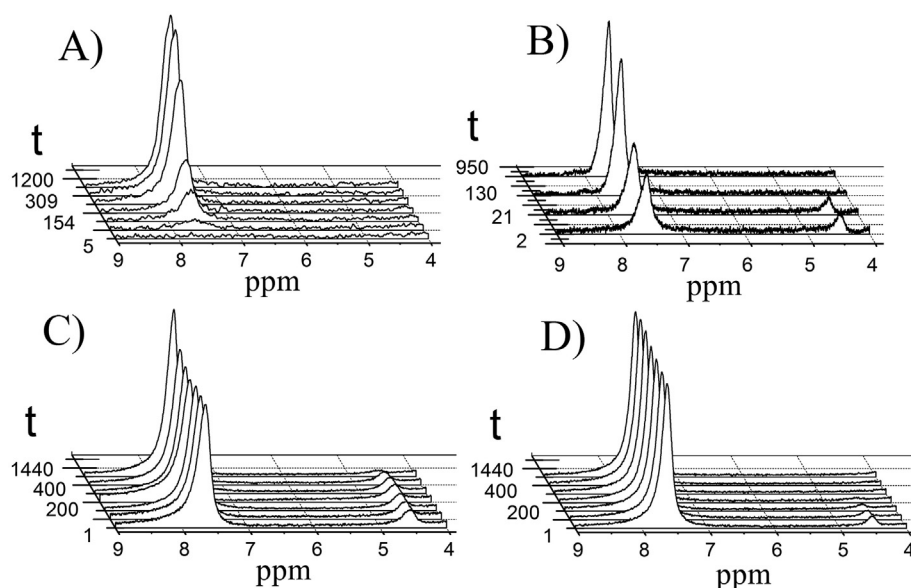
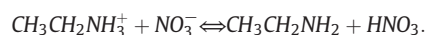


Fig. 6. Changes of $-\text{NH}_3$ and H_2O signals in ^1H NMR spectra of EAN- H_2O mixtures with: A) 0, B) 3.1, C) 6.2 and D) 12.4 mol% of added H_2O at different times (minutes) of exposure of the sample to a magnetic field of 9.4 T. Spectra were acquired by Fourier transformation of spin echo (SE) at $\tau = 5.634$ ms, $g = 0.006$ T/m, $\delta = 1$ ms.

intensities of the resonance lines is due to changes in the NMR relaxation times for the $-\text{NH}_3$ and H_2O protons. For protons of the $-\text{NH}_3$ groups in confined EAN without added water, the mechanism of proton “exchange” has been suggested and discussed earlier [21]. $-\text{NH}_3$ protons are delocalized between the $-\text{NH}_3$ group of the EA^+ cation and NO_3^- anion and their exchange is described by the following equilibrium [32]:



In the bulk EAN, protons of $-\text{NH}_3$ of the EA cation yield a narrow line [21]. Therefore, they are in a state of “fast exchange”, where the difference in the chemical shift frequencies of the two sites is much less than the exchange rate [33]. This is a case of the so-called “motional averaging” exchange regime. For EAN placed between polar glass plates, the exchange rate decreases and the $-\text{NH}_3$ protons are in the “intermediate exchange” regime where the difference in the chemical shift frequencies of the two sites is of the order of the exchange rate. If this is so, from Fig. 6 it followed that increasing the water content leads to slowing of the exchange rate and some weakening of the effect of the magnetic field on alteration of the exchange during exposure of the sample to the magnetic field. From our current experiments presented in Figs. 5 and 6, it is evident that water plays a crucial role in the exchange processes, even if it is residual water in a confined EAN sample prepared without added water (Fig. 6A). The decrease in the signal of water after exposure to the magnetic field leads to recovery of a sharp line of the $-\text{NH}_3$ group.

Possible mechanisms leading to shortening the ^1H NMR relaxation of water protons of the residual water during long exposure to the magnetic field in EAN without added water have been discussed and analyzed earlier [24]. Using presaturation experiments in ^1H NMR spectroscopy, we demonstrated that proton exchange is not the reason for the decreased T_2 NMR relaxation time. It was also suggested that the most probable mechanism for the decrease of the bulk water signal is adsorption of water on polar surfaces of glass plates [24]. It is known that adsorbed water possesses “solid-like” properties [34] and, therefore, does not contribute to the ^1H NMR echo signal of a highly mobile liquid phase.

From our experimental data, it is seen that confinement of the EAN-water mixture between polar glass plates has effects on the diffusivity of both, the EA cation, and water. Exposure to a static magnetic field affects the diffusivity of EAN, but not the apparent (measured) diffusivity of

water. Because amplitude of the water signal decreases during the exposure, we can suggest that water is present in two states with different dynamic properties. One type of water formed in confinement before the application of a static magnetic field possesses a long T_2 relaxation time typical for liquid and diffusivity measured by ^1H NMR. The second type of water is formed during exposure of the sample of the first type to the magnetic field and finally includes all water present in the system. This type of water possesses “solid-like” NMR relaxation that makes it “invisible” to the NMR diffusometry technique. We suggest that this second type of water is surface adsorbed water. Correspondingly, EAN is contained in two liquid phases: the first one contains EAN-water mixture, while the second one, which forms under the influence of the static magnetic field, contains neat EAN. Generally, this agrees with the earlier suggestion of Atkin and Warr about the possible molecular exfoliation of ILs placed between two plates [35]. We suggest that exposure to a static magnetic field leads to reversible adsorption-desorption of water on/from surfaces of polar glass plates in the confined EAN-water mixture. In the process of the phase separation, NMR spectroscopy detected signals from both the neat EAN and the EAN-water phases. NMR spectra do not contain an information concerning the scale range of the process, while the change of diffusivity of water is informative. The presence of nano droplets of the phase containing water, restricted diffusion of water would lead to decrease its apparent diffusion coefficient measured by NMR. This decrease does not take place in our experiment; diffusion coefficient of water does not change in the course of phase separation. Therefore, the process occurs in the micrometre scale range of the whole water layers present between the glass plates. This agrees with Avrami kinetics of the phase transformation when a new phase grow on the previously formed nucleus and develops through attainable space range.

Thicknesses of the water layers on the glass surface formed after the repealing of water from the EAN-water mixture can be estimated from a simple geometrical consideration. Between each two plates two layers of water and one layer of EAN are formed. The ratio of water and EAN layers is:

$$l_{\text{H}_2\text{O}}/l_{\text{EAN}} = 0.5 \cdot (m_{\text{H}_2\text{O}}/m_{\text{EAN}}) \cdot (\rho_{\text{EAN}}/\rho_{\text{H}_2\text{O}})$$

where $m_{\text{H}_2\text{O}}$ and m_{EAN} are masses in the layer, and ρ_{EAN} and $\rho_{\text{H}_2\text{O}}$ are densities of EAN and water, respectively. Taking in to account the density of EAN 1.261 g/ml and mean spacing between glass plates ca. 4

μm , thicknesses of layers of water in the EAN-water systems confined between glass plates are: 12.6 nm (3.1 mol% H_2O), 25 nm (6.2 mol% H_2O) and 50 nm (12.4 mol% H_2O). Thicknesses of water layers are of the order of values of roughness of the microscope glass plates surfaces that may contain inhomogeneities up to 4 nm in height [36]. This might be a cause for a local inhomogeneity of the magnetic field and additionally contribute to T_2 NMR relaxation of adsorbed water.

In previous studies of bulk EAN-water mixtures, it was suggested that water molecules at low water concentrations may be included in the network of intermolecular bonds of EAN forming homogeneous solutions [8]. This agrees with the diffusion behaviour of EAN and water in the EAN-water mixture confined between polar glass plates before exposure to the static magnetic field, because their diffusion coefficients exhibit similar trends with increased water concentration. However, this is not the case for the system subjected to the magnetic field, where EAN and water demonstrate different trends during the time of exposure.

4. Conclusions

Diffusion in EAN-water mixtures confined between parallel polar glass plates demonstrates opposite trends at increased water concentrations in comparison with the bulk mixture: the diffusivities of the ethylammonium cation and water decrease with increasing concentration of water. For the system exposed to a static magnetic field, the diffusion of the EA cation decreased, a process described by the Avrami equation typical for phase transitions. At the same time, the measured diffusion coefficient of water remains unchanged with the ^1H NMR signal of water gradually decreasing to zero. We suggest that a static magnetic field has the effect of gradually decomposing the EAN-water phase into a neat EAN liquid phase and a “solid-like” water phase adsorbed onto the polar surface of glass plates. The presence of water is crucial for exchange of protons of the $-\text{NH}_3$ group of EAN.

Behavior of NO_3^- anion of the EAN ionic liquid remains not clear. To illuminate this issue, we suggest experiments with EAN containing ^{15}N and ^{17}O enriched NO_3 for further studies.

We demonstrated how confinement and exposure to a static magnetic field is important for the dynamics and phase state of a mixture of a protic ionic liquid with water. We suggest that the results will have strong implications for interface-intensive applications of ionic liquids, such as their use in machinery, electrochemical and electro-magnetic systems to develop new methods to manipulate the properties of ionic liquids by magnetic fields for future applications in lubrication, electrical storage devices, supported organic catalysis and portable electronic gadgets.

Acknowledgments

This research project did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors. We are grateful to the Kempe Foundation in memory of J.C. and Seth M. Kempe and the LTU laboratory fund, which provided a few grants that funded the purchase of the NMR equipment. NMR measurements were partly carried out on equipment of the Federal Centre of Collective Facilities of Kazan Federal University. “Scriptia Academic Editing” is acknowledged for English correction and proofreading of this manuscript.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.molliq.2018.10.080>.

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